A continuous parametrisation of phylogenetic trees and application for Bayesian phylogenetic inference

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1 Introduction

We consider phylogenetic trees that are binary and rooted, with n labeled leaves. There are $n-1$ internal nodes including the root. Such trees are usually represented as a mixture of discrete and continuous variables (Paradis 2006). The topology of the tree is discrete and there are $(2n-3)!! = \frac{(2n-2)!}{2^{n-1}(n-1)!}$ unique possibilities (Felsenstein 1978). The branch lengths are continuous, and there are $2n-2$ branch lengths taking values in $\mathbb{R}_{>0}$ (one for each leaf and non-root internal node). This mixture makes it difficult to sample the space of trees. For example, Markov Chain Monte-Carlo (MCMC) methods (Gilks et al. 1996) need to propose updates to both the branch lengths and the topology, using moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). These moves have been the workhorse of Bayesian phylogenetics for the past 30 years but are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020). Alternatives such as Hamiltonian Monte-Carlo (HMC) can only deal with continuous variables (Betancourt 2017). A continuous representation of phylogenetic tree is given by the Billera-Holmes-Vogtmann (BHV) space (Billera et al. 2001), which is made of one orthant for each topology, connected between themselves at the points where a branch length degenerates to zero to switch from one topology to another. But this construction is difficult to use in Bayesian phylogenetic inference.

2 New continuous parametrisation

Here we propose a new continuous parametrisation for binary rooted phylogenetic trees. We consider an arbitrary but fixed order in which the leaves are added to the tree. Let L_{i-1} denote the sum of the branch lengths in the tree formed by the first $i-1$ leaves. For the first leaf ($i=1$) there is nothing to specify but from the second leaf ($i = 2..n$) two parameters need to be specified, a_i and b_i which both take values in $\mathbb{R}$. The parameter a_i determines the point on the tree built so far where the new leaf is grafted. The parameter b_i determines the length $\exp(b_i)$ of the branch separating the new leaf from this grafting point. The parameter a_i represents the length of the path from the root of the tree to the grafting point, following a contour traversal started at the root of the tree. The contour traversal algorithm is defined recursively as follows: start from a given point, move down to its first child, performs the contour traversal from this first child, then move back up from the first child to the starting point, then move down to the second child, perform the contour traversal from the second child, and then move back up to the starting point. If the contour traversal comes back to the root after having explored both children, which happens if and only if $a_i > 2L_{i-1}$, then the grafting point

is above the root by the remaining amount $a_i - 2L_{i-1}$. Likewise, if $a_i < 0$ then the grafting point is above the root by the amount $-a_i$. We can deduce the following recursive formula for the sum of the branch lengths:

$$L_i = \begin{cases} 0 & \text{if } i = 1 \\ L_{i-1} + \exp(b_i) + \max(0, a_i - 2L_{i-1}) + \max(0, -a_i) & \text{otherwise} \end{cases} \tag{1}$$

The parametrisation of a_i using a contour traversal is illustrated in Figure 1. At the graft point, a new node is created with left child equal to what was previously below that point, and right child equal to the new branch of length $\exp(b_i)$ leading to the leaf number i. This process is repeated for all leaves from the second one ($i = 2$) to the last one ($i = n$), each time using the contour traversal on the tree built previously.

This parametrisation has the advantage to create no discontinuity, or no jump from one part of the tree to another. It has the disadvantage to represent each grafting point exactly twice, since each point of the tree is traversed first as the algorithm goes down the branch and then again as it goes back up the branch. This is not ideal but is the price to pay to achieve continuity, and we will see that it is not a problem in practice. Let $C = (a_i, b_i)_{i=2..n} \in \mathbb{R}^{2n-2}$. C defines a unique phylogenetic tree T, which can be reconstructed by considering adding the leaves one by one as described above. Conversely, each unique phylogenetic tree T can be obtained by several values C due to each grafting point having two equivalent values of a_i. There are therefore 2^{n-1} unique values of C for each phylogenetic tree T.

3 Sampling the continuous tree space with MCMC

We consider a prior distribution on the tree T such that all of the $(2n-3)!!$ topologies are equally likely and the branch lengths l_i follow a Gamma(α, θ) distribution, so that:

$$p_T(T) = \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \frac{1}{\Gamma(\alpha)\theta^\alpha} l_i^{\alpha-1} \exp\left(-\frac{l_i}{\theta}\right) \tag{2}$$

To find the corresponding distribution on C we need to consider the Jacobian of the function $f : C \to T$:

$$\left|\det \frac{\partial f(C)}{\partial C}\right| = \exp\left(\sum_{i=2}^{n} b_i\right) \tag{3}$$

When accounting for this and the multiple representations of each T in the space of C we obtain the probability density function of T:

$$p_C(C) = \frac{1}{2^{n-1}} p_T(f(C)) \exp\left(\sum_{i=2}^{n} b_i\right) \tag{4}$$

To sample from this distribution, we use the adaptive Metropolis sampler (Vihola 2011). This algorithm uses a multivariate normal distribution as the proposal kernel $q(C'|C)$ which is symmetric ($q(C'|C) = q(C|C')$), so that the acceptance probability is simply $\min(1, p_C(C')/p_C(C))$ calculated using Equation 4. The mean of the proposal kernel is the current value C and its covariance matrix is tuned to achieve an acceptance rate of 0.234 (Andrieu and Thoms 2008). We take $n = 5$ as a small example, with $\alpha = 2$ and $\theta = 2$ as the shape and scale parameters of the distribution of branch lengths. The sampler was run for 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 100 iterations to produce a sample of size 1000. This run took less than a minute on a standard laptop. Figure 2 shows that the trees sampled by the MCMC are distributed as expected under Equation 2. All $(2n-3)!! = 105$ topologies have been sampled, the distribution of the sum of branch lengths and of a single branch length are as expected, and a majority-rule consensus tree looks completely star-like. The effective sample size was greater than 200 for individual branch lengths as computed using the R package coda (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 200 as computed by the R package rwty (Warren et al. 2017).

4 Sampling the continuous tree space with HMC

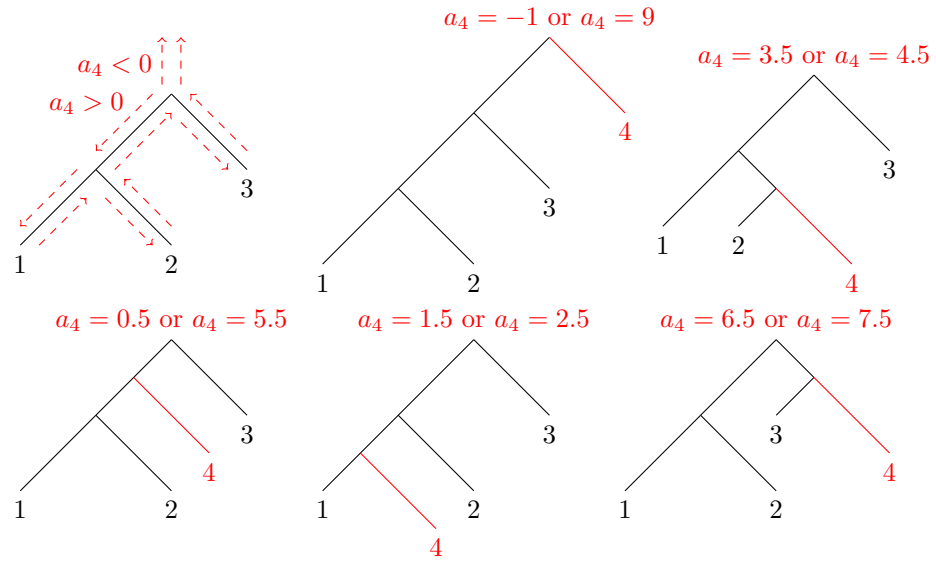


Figure 1: Illustration of the contour traversal. The initial tree is shown in black, with three leaves labeled 1, 2 and 3 and all four branches of length 1. Leaf 4 is added with the value of a_4 determining the graft point as shown and the new terminal branch of length $\exp(b_4) = 1$ shown in red.

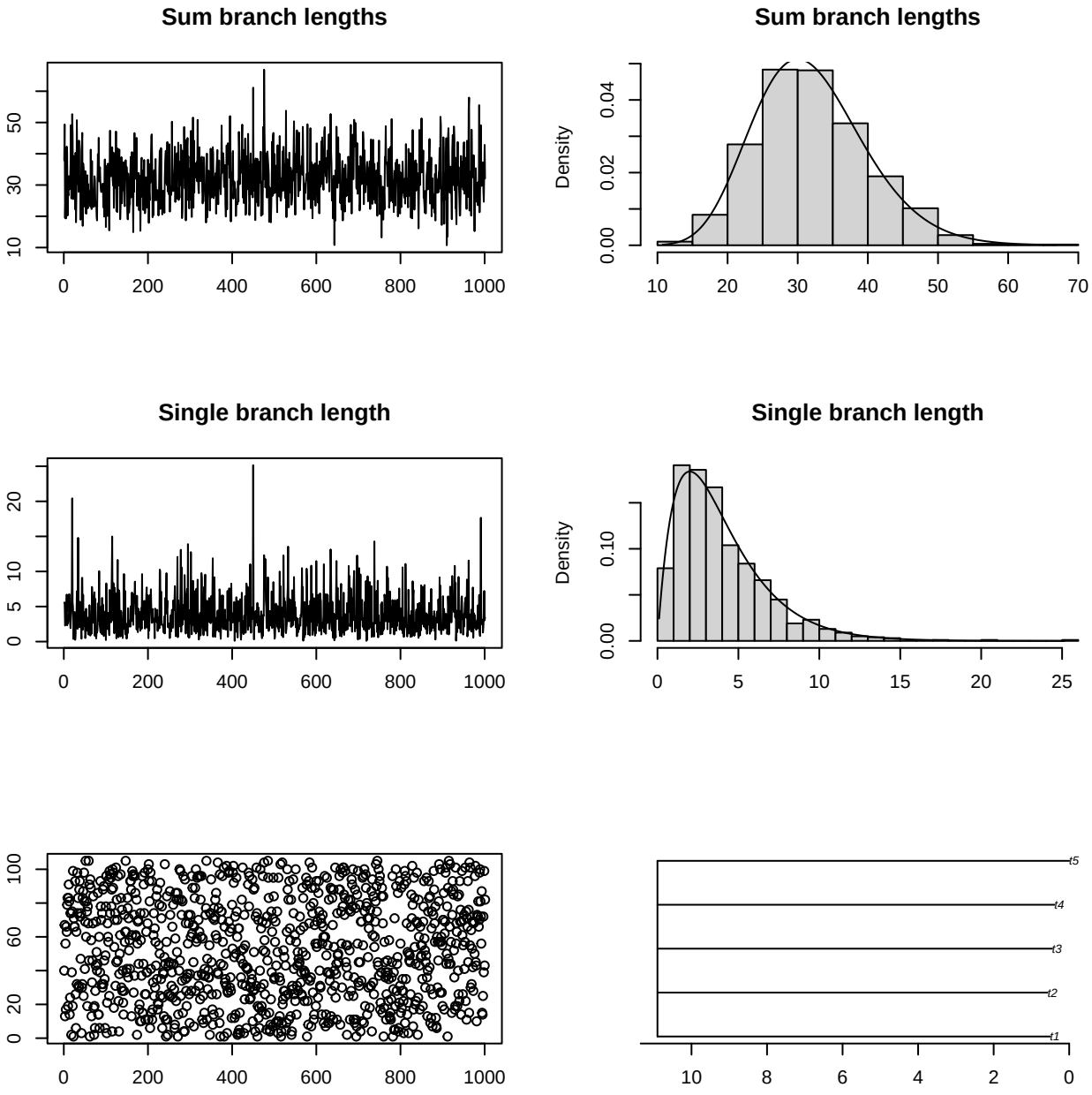


Figure 2: MCMC run. (A) Trace of sum of branch lengths. (B) Distribution of sum of branch lengths, sampled (histogram) and expected (line). (C) Trace of a single branch length. (D) Distribution of a single branch length, sampled (histogram) and expected (line). (E) Trace of topology. (F) Majority-rule consensus tree.

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